Assignment 1: Markdown Buddy

Drafting and Formatting R Documentation with Generative AI — Reflection Answers

AI ASSISTANCE DECLARATION

AI Assistance Declaration: I utilized ChatGPT to assist in structuring, drafting, and organizing these reflection responses regarding AI-guided README creation and R Markdown documentation. Prompt sequences focused on analyzing AI limitations, establishing verification protocols, and promoting ethical usage guidelines. I manually verified all arguments and statements against the actual repository structure and underlying R scripts. All final checks and calculations are conducted by myself, and I take full responsibility for the originality and technical accuracy of this work.

1. How did AI assist in creating and improving the R project documentation?

Artificial intelligence served as an efficient structural and drafting assistant throughout the synthetic retail sales analysis task. It proposed a cohesive README layout, logically sequencing elements from an initial project overview down to software prerequisites, setup instructions, executable commands, expected outputs, model constraints, and citations. Establishing this standardized outline is vital, as effective documentation must enable peers and external collaborators to comprehend and replicate the analytical pipeline independently without direct verbal guidance. Additionally, the AI recommended clean header hierarchies and code-block formatting within the R Markdown document, ensuring clear boundaries between the script's primary objective, raw data inputs, execution routines, and generated outputs.

The primary advantage of incorporating AI lay in the iterative drafting process, allowing continuous refinement through targeted prompt sequences rather than composing all structural elements manually. For instance, an initial prompt requesting core README sections was subsequently refined to request concise Markdown header tags, followed by a critique targeting visual flow and syntax clarity. Nonetheless, AI suggestions were treated strictly as preliminary drafts rather than authoritative truth; all directory file paths, software library dependencies, and terminal commands still required manual alignment against the physical repository files prior to submission.

2. What challenges and limitations did you identify when reviewing AI-generated Markdown?

A primary limitation discovered during review is that polished, well-written prose can easily mask underlying technical errors. An AI-crafted README might cite non-existent software packages, reference output files that the code never generates, or output invalid relative directory paths. Furthermore, Markdown structures such as nested bullet lists, complex headers, and fenced code blocks may look readable in plain text but fail or misalign when rendered in GitHub or RStudio previewers. Catching these inaccuracies is essential because end-users rely entirely on documentation accuracy to execute the workflow.

To address these challenges, the established verification workflow involves cross-referencing every documented instruction against the physical R code, confirming that all datasets remain synthetic, and verifying the visually rendered README directly on GitHub. The R Markdown source file must be compiled (knitted) within RStudio or Posit Cloud to confirm that code chunks execute seamlessly and that graphics and data tables render properly. Numerical outputs must be cross-checked against raw records rather than accepted purely on AI summary assertions. Finally, a thorough manual review ensures uniform header styling, working internal links, and clear explanations—serving as an active verification protocol rather than an unverified assumption of accuracy.

3. What did you learn about ethical AI use and how would you improve this workflow?

The core takeaway from this project is that while AI tools can accelerate technical writing and enhance structural clarity, full accountability remains squarely with the human author. Ethical usage mandates transparent disclosure of the AI tool employed, detailed records of prompts and major edits, and a clear distinction between AI-assisted text generation and independently calculated findings. Additionally, limiting inputs to synthetic datasets ensures strict data privacy and prevents sensitive or proprietary data from leaking into public model prompts or online repositories.

An optimized future workflow would start with a fully tested R script and a manually authored summary of inputs and expected outputs. AI would then be leveraged solely to suggest clean formatting structures and README outlines, after which every generated section would be validated directly against the working code. Next, I would render the markdown preview, compile the R Markdown report, audit sample data calculations, and refine ambiguous phrasing. Lastly, I would attach a concise prompt history appendix alongside a clear summary of completed validation steps—using AI strictly as a drafting aid while taking full personal ownership of reproducibility and technical integrity.